

MICHAEL GARCIA

Data Engineer | PhD in Physics | Modern Data Stack Expert

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À PROPOS

PhD in Physics & Data Engineer. I don't just build pipelines; I craft financial assets. Specializing in the Modern Data Stack, I translate technical complexity into immediate competitive advantage and business value.

FORMATION

- 2026 **Data Engineering Zoomcamp (200h Training)** – DataTalks.Club
Advanced specialization: Batch & streaming pipeline architecture. Mastery of Cloud ecosystem (AWS/GCP), dbt, Snowflake, Spark, Kafka, and Terraform (IaC). Orchestration with Kestra.
- 2024 **Data Engineering Intensive Training (400h Training)** – Data-Upskilling
Software engineering applied to Data: Python (production-ready), SQL expert, Orchestration (Airflow), Git, and CI/CD workflows.
- 2013 **PhD in Physics for Life Sciences** – Université Grenoble Alpes

COMPÉTENCES TECHNIQUES

Data Architecture & Platforms

- Data Warehousing: Snowflake, BigQuery, PostgreSQL, DuckDB
- Transformation & Modeling: dbt Core, SQL expert
- Orchestration & Ingestion: Airflow, Kestra, dlt

Cloud & Software Engineering

- Infrastructure: GCP, AWS, Azure, Terraform, Docker
- Development: Python (Pandas, Polars, PySpark, Kafka), Scraping, TypeScript
- DevOps & Quality: CI/CD (GitHub Actions), Unit Testing (Pytest), Monitoring (Grafana)

EXPÉRIENCE

- 05/2026-06/2026 **Consultant Data Engineer (Freelance)** – Alpine Data - client: Wetall (E-commerce & Affiliation)
- Revenue protection through automation
- 📊 **Audit & Consulting** – Analysis of conversion losses and deployment of an automated product availability verification pipeline.
 - 🔗 **Data Ingestion** – Design of a scraping and API integration system to synchronize partner catalogs.
 - 🔧 **Stack** : Python, Web Scraping, API Integration, GitHub Actions, Automation
 - 🌐 **Wetall Website**
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- 2026 **Data Engineer & Analyst** – Energy Risk for SME Project
- Modeling SME exposure to electricity price volatility
- 📊 **Business Impact** – P&L impact simulation (e.g., -38% gross margin) and visualization via Looker Studio.
 - 🏗️ **FinOps Architecture** – Ingestion pipeline to BigQuery. Cost and performance optimization via partitioning and clustering.
 - 🔗 **dbt Pipeline** – OPSD data transformation: Unpivot and dynamic Cross Join with business segments (Bakery, Retail).

🔧 *Stack : GCP, BigQuery, dbt Core, Terraform, Python, Looker Studio*

🔗 [GitHub Code](#) · [Looker Dashboard](#)

2026

Data Engineer (Alpine Lambda Trade Project) – [Market Intelligence](#)

Real-time analysis and volatility alerts (BTC/ETH)

🔧 *Hybrid Architecture* – Design of a Lambda architecture (Batch + Streaming) for processing live Bitcoin and Ethereum prices on Snowflake.

⚙️ *Ingestion & Monitoring* – Bollinger Bands analysis to detect rapid variations and trigger automatic alerts (sharp surges/drops).

🔧 *Stack : Snowflake, AWS, Kafka, Python, Terraform, dbt, Kestra*

🔗 [GitHub Code](#)

2026

Cloud Data Engineer (Grade-Scale Project) – [AI Evaluation Engine & Cloud Infrastructure](#)

Automated Azure Cloud architecture and AI agents

🔧 *Infrastructure as Code* – Automated deployment on Azure via Terraform to ensure environment reproducibility.

💻 *Backend & AI* – Design of a scalable backend integrating intelligent analysis via LLMs (Groq).

🔧 *Stack : Terraform, Azure, NodeJS, TypeScript, Zod, Groq LLM*

🔗 [GitHub Code](#) · [Live Demo](#)

2013 – 2026

Teacher / Software Developer – [French Ministry of National Education](#)

Project Management & Pedagogy

💻 *Translation* – Needs analysis and automation of administrative processes using custom Python tools.

👥 *Knowledge Transfer* – Simplification of complex technical concepts for non-expert audiences.

2009 – 2013

PhD Researcher – [LIPhy - CNRS Grenoble](#)

Data Engineering & Modeling

🔧 *ETL Pipelines* – Automation of the imaging pipeline, reducing analysis time by 50%.

📊 *Data Quality* – Algorithms for detection, reconstruction, and cleaning of trajectories (Bronze/Silver/Gold).

🔧 *Stack : Python, IDL (matlab-like), Image Processing, Data Modeling*

LANGUES

French:

Native

English:

Professional (C1 - Fluent)